Virtualizaton & Cloud Virtual Machines

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Learning Objectives

- Explain what virtualization is and why it is the foundational technology behind cloud computing.
- Distinguish between Type 1 and Type 2 hypervisors and identify which type cloud providers use.
- Compare full virtualization, paravirtualization, and hardware-assisted virtualization.
- Describe how AWS, GCP, and Azure implement virtualization and hardware offloading in their infrastructure.
- Explain the relationship between virtualization and Infrastructure as a Service (IaaS), including how EC2 and GCE expose virtualized compute.
- Evaluate and select appropriate instance/machine types, pricing models, and tenancy options for a given workload.
- Describe the role of machine images (AMIs), root device volumes, and instance metadata in the VM lifecycle.
- Explain the Meltdown and Spectre vulnerabilities and articulate why hardware-level vulnerabilities pose unique risks to multi-tenant cloud environments.
- Evaluate the security tradeoffs of running workloads in a shared cloud infrastructure, including the role of security groups, dedicated hosts, and confidential computing.

Foundations of Virtualization

Limitations of Bare-Metal (Physical) Servers

- Traditional servers are designed to run one operating system and one application at a time
- This leads to underutilization — most servers run at 10–15% of capacity
- Hardware refresh cycles are slow; provisioning new physical servers takes weeks to months

What Is Virtualization?

- In computing, virtualization refers to the act of creating a virtual (rather than actual) version of something, including virtual computer hardware platforms, storage devices, and computer network resources
- The core idea: using software to simulate the existence of hardware and create virtual computer systems
- This allows businesses to run more than one virtual system — and multiple operating systems and applications — on a single server, providing economies of scale and greater efficiency

How Cloud VMs Work

The Virtual Machine

- A virtual computer system is known as a "virtual machine" (VM): a tightly isolated software container with an operating system and application inside
- Each self-contained VM is completely independent
- Putting multiple VMs on a single computer enables several operating systems and applications to run on just one physical server, or "host"

The Hypervisor

A thin layer of software called a hypervisor decouples the virtual machines from the host and dynamically allocates computing resources to each virtual machine as needed

- **Type 1 (Bare-Metal):** Runs directly on hardware — used by all major cloud providers (KVM, Xen, Hyper-V, VMware ESXi)
- **Type 2 (Hosted):** Runs on top of a host operating system — used for local development (VirtualBox, VMware Workstation, Parallels)

Key Properties of Virtual Machines

- **Partitioning:** Run multiple operating systems on one physical machine; divide system resources between virtual machines
- **Isolation:** Provide fault and security isolation at the hardware level; preserve performance with advanced resource controls
- **Encapsulation:** Save the entire state of a virtual machine to files; move and copy VMs as easily as moving and copying files
- **Hardware Independence:** Provision or migrate any virtual machine to any physical server

Server Consolidation

- Using server virtualization, a company can maximize the use of its server resources and reduce the number of servers required
- The result is server consolidation, which improves efficiency and cuts costs

Virtualization Techniques

- **Full Virtualization:** Guest OS is completely unaware it is virtualized; hypervisor translates all hardware calls. Simpler for guests but historically slower.
- **Paravirtualization:** Guest OS is modified to communicate directly with the hypervisor via "hypercalls." Better performance but requires kernel modification. (Xen PV is a classic example.)
- **Hardware-Assisted Virtualization:** CPU provides native support for virtualization (Intel VT-x, AMD-V). Enables near-native performance without guest OS modification. This is the dominant approach in modern cloud infrastructure.

Virtualization Is Not Cloud Computing

- Cloud computing is not the same thing as virtualization; rather, it is something you can do using virtualization
- Cloud computing describes the delivery of shared computing resources (software and/or data) on demand through the Internet
- Virtualization is the enabling technology; cloud computing is the delivery model built on top of it

VMs vs. Containers

- Virtual machines virtualize the hardware; containers virtualize the operating system
- VMs provide strong isolation via the hypervisor; containers share the host kernel
- Cloud computing uses both: VMs for infrastructure isolation, containers (Docker, Kubernetes) for application packaging
- The tradeoff: VMs offer stronger security boundaries; containers offer faster startup and lower overhead

Virtualization as IaaS

EC2 and Google Compute Engine

From Virtualization to Infrastructure as a Service

- Virtualization enables IaaS — the cloud service model where the provider manages the physical infrastructure and the customer manages everything from the OS up
- Amazon EC2 and Google Compute Engine are the IaaS compute services that expose virtualized hardware to customers

Amazon EC2

- Amazon Elastic Compute Cloud (Amazon EC2) is a web service that provides secure, resizable compute capacity in the cloud
- EC2 reduces the time required to obtain and boot new server instances to minutes
- Pay only for capacity that you actually use
- **Amazon EC2 is a virtual machine in the AWS cloud platform**

Google Compute Engine

- Compute Engine instances can run public images for Linux and Windows Server that Google provides, as well as private custom images that you can create or import from your existing systems
- Regardless of the region where you create your VM instance, the default time for your VM instance is Coordinated Universal Time (UTC)

Benefits of VMs in Cloud

- Elastic computing, complete control, flexibility, reliability, security, pay-as-you-go, integrated services, and easy to start

Instance and Machine Types

- AWS and GCP provide a wide selection of instance types optimized to fit different use cases
- Instance types comprise varying combinations of CPU, memory, storage, and networking capacity
- Each instance type includes one or more instance sizes, allowing you to scale resources to the requirements of your target workload

AWS EC2 Instance Type Categories

- General Purpose — balanced compute, memory, and networking (e.g., t3, m6i)
- Compute Optimized — high-performance processors for compute-intensive workloads (e.g., c6i)
- Memory Optimized — fast performance for workloads that process large data sets in memory (e.g., r6i)
- Accelerated Computing (GPU) — hardware accelerators for ML, HPC (e.g., p4d, g5)
- Storage Optimized — high sequential read/write access to large datasets on local storage (e.g., i3, d3)

GCE Machine Type Families

- General-purpose — best price-performance ratio for a variety of workloads
- Compute-optimized — highest performance per core, optimized for compute-intensive workloads
- Memory-optimized — ideal for memory-intensive workloads, with up to 12 TB of memory
- Accelerator-optimized — ideal for massively parallelized CUDA compute workloads such as ML and HPC

Tenancy: Dedicated Instances vs. Dedicated Hosts

- **Dedicated Instances (AWS):**
  - EC2 instances that run in a VPC on hardware dedicated to a single customer
  - Physically isolated at the host hardware level from instances that belong to other AWS accounts
  - May share hardware with other instances from the same AWS account that are not Dedicated instances
  - Per-instance billing (subject to a $2 per region fee)
- **Dedicated Hosts (AWS):**
  - A physical EC2 server dedicated for your use
  - Help reduce costs by allowing you to use your existing server-bound software licenses (Windows Server, SQL Server, SUSE Linux Enterprise Server)
  - Help meet compliance requirements
  - Per-host billing
  - Provide visibility of sockets, cores, and host ID
  - Support affinity between a host and instance, targeted instance placement, and capacity allocation requests

Tenancy: Dedicated Instances vs. Dedicated Hosts

- **GCP Equivalent:** Sole-Tenant Nodes — dedicated physical servers for your workloads only
- **Azure Equivalent:** Azure Dedicated Hosts — physical servers dedicated to your organization

Pricing Models

AWS EC2 Pricing Models

- **Billing Granularity:** All vCPUs, GPUs, and GB of memory are charged a minimum of 1 minute. After 1 minute, instances are charged in 1-second increments.
- **On-Demand:** Pay for compute capacity by the hour with no long-term commitments or upfront payments. Recommended for short-term, spiky, or unpredictable workloads that cannot be interrupted, and for applications being developed or tested for the first time.
- **Reserved Instances:** Significant discount (up to 75%) compared to On-Demand pricing. Requires 1- or 3-year commitment. Recommended for applications with steady-state usage that may require reserved capacity.
- **Spot Instances:** Bid on spare Amazon EC2 computing capacity for up to 90% off the On-Demand price. Recommended for applications that have flexible start and end times, are only feasible at very low compute prices, or need large amounts of additional capacity urgently. Instances can be interrupted.
- **Dedicated Hosts:** Per-host billing with visibility into physical hardware. Useful for bring-your-own-license (BYOL) scenarios.
- **Dedicated Instances:** Per-instance billing on hardware dedicated to a single customer.

GCE Pricing Model

- **Billing Granularity:** All vCPUs, GPUs, and GB of memory are charged a minimum of 1 minute. After 1 minute, instances are charged in 1-second increments.
- **Spot VMs (Preemptible VMs):** Automatic discount between 60–91%. Spot prices change over time but discounts are always in this range. Equivalent to AWS Spot Instances.
- **Committed Use Discounts (CUDs):** Up to 70% discount for memory-optimized machine types, up to 57% for all other machine types. Requires 1- or 3-year commitment. Equivalent to AWS Reserved Instances.
- **Sustained Use Discounts (SUDs):** Automatic discount of up to 30% on resources used for more than 25% of a month and not receiving any other discounts. No commitment required — applied automatically.
- **Important:** Discount types cannot be combined. Spot VMs cannot receive sustained use discounts or CUDs.

Azure Pricing Model

- **Pay-as-you-go:** Similar to AWS On-Demand.
- **Azure Reserved VM Instances:** 1- or 3-year commitments for up to 72% savings.
- **Azure Spot VMs:** Similar to AWS Spot Instances.
- **Azure Savings Plans:** Flexible commitment-based discounts across multiple services.

Cloud VM Features and Configuration

Security Groups

- A security group acts as a virtual firewall that controls the traffic for one or more instances
- When you launch an instance, you associate one or more security groups with the instance
- You add rules to each security group that allow traffic to or from its associated instances
- Rules can be modified at any time; new rules are automatically applied to all associated instances
- All rules from all security groups associated with an instance are evaluated to decide if traffic should be allowed
- GCP equivalent: Firewall Rules; Azure equivalent: Network Security Groups (NSGs)

Instance Metadata and User Data

- Instance metadata is data about your instance that you can use to configure or manage the running instance
- You can use instance metadata to access user data specified when launching your instance
- User data can include parameters for configuring your instance or a startup script
- AWS: http://169.254.169.254/latest/meta-data/
- GCP: http://metadata.google.internal/computeMetadata/v1/

Other Cloud VM Features

- Secure login using key pairs (SSH keys)
- Temporary and permanent storage volumes
- Static IPv4 addresses (AWS: Elastic IP; GCP: Static External IP; Azure: Public IP)
- Metadata tags for organization and automation
- Subnets (virtual networks) for network segmentation

Meltdown, Spectre, and the Security of Shared Hardware

Background: How CPUs Optimize Performance

- **Out-of-Order Execution:** Modern CPUs do not execute instructions strictly in order. They look ahead and execute instructions that are ready, rearranging results later.
- **Speculative Execution:** CPUs predict which branch of a conditional statement (e.g., an if/else) will be taken and begin executing it before the condition is resolved. If the prediction is wrong, the results are discarded — but traces remain in the CPU cache.
- **CPU Cache and Side Channels:** The cache is faster than main memory. By measuring how long it takes to access data, an attacker can infer what data is in the cache — even data they should not have access to. This is called a **side-channel attack.**

Meltdown (CVE-2017-5754)

- Disclosed January 2018 by Google Project Zero and independent researchers
- Exploits out-of-order execution on Intel processors (and some Arm cores). AMD processors were not affected.
- Allows a user-space program to read kernel memory, breaking the fundamental isolation between applications and the operating system
- The CPU speculatively accesses privileged memory, and although the access is eventually blocked, the data briefly enters the cache. A side-channel timing attack can then extract it.
- **Impact on cloud:** In a cloud environment, this means a VM could potentially read data from the hypervisor or host kernel memory, which may contain data from other tenants.
- **Mitigation:** Kernel Page-Table Isolation (KPTI) — separates kernel and user-space memory mappings. Effective but introduced a performance penalty of 5–30% for I/O-intensive workloads.

Spectre (CVE-2017-5753 and CVE-2017-5715)

- Also disclosed January 2018
- Affects virtually all modern processors: Intel, AMD, and Arm
- Exploits speculative execution to trick a program into leaking its own data. Unlike Meltdown, Spectre targets the boundaries between applications rather than between the application and the kernel.
- Two main variants:
  - **Variant 1 (Bounds Check Bypass):** Exploits speculative execution past array bounds checks.
  - **Variant 2 (Branch Target Injection):** Manipulates the CPU's branch predictor to speculatively execute attacker-chosen code in another process's context.
- Spectre is harder to exploit than Meltdown but also significantly harder to mitigate because it is embedded in CPU architecture rather than a specific implementation flaw.
- **Mitigation:** A combination of microcode updates, compiler-level mitigations (e.g., Google's "Retpoline" for Variant 2), and OS patches.

Impact on Public Cloud — Why This Was a Crisis

- **The core problem:** Cloud computing is built on multi-tenancy — multiple customers share the same physical CPU. Meltdown and Spectre broke the assumption that one tenant's data is invisible to another.
- **Specific cloud risks:**
  - A malicious VM on shared hardware could read host kernel memory (Meltdown) or data from co-located VMs (Spectre)
  - Paravirtualized environments (Xen PV) were especially vulnerable because they share the same kernel
  - Containers sharing a kernel (Docker, LXC) were at higher risk than hardware-virtualized VMs
- **Cloud provider response:**
  - **AWS:** Patched the entire EC2 fleet, with only a small percentage of instances requiring reboots. AWS credited its Nitro architecture for limiting exposure since Nitro's minimal hypervisor has a much smaller attack surface than the old Xen Dom0.
  - **GCP:** Used live migration to patch hosts without customer-visible reboots. Google's Project Zero team had actually discovered the vulnerabilities. Google also developed Retpoline, a software mitigation for Spectre Variant 2.
  - **Azure:** Applied patches and mitigations, though some customers experienced performance degradation. Microsoft released microcode updates and coordinated guest OS patching.
- **Performance impact:** Patches introduced measurable slowdowns. Red Hat measured 1–20% impact. Epic Games reported a ~300% increase in backend CPU utilization on AWS after patching. I/O-heavy workloads suffered the most.

Risks to Your Compute Workloads

- **Data exfiltration from co-tenants:** In a shared environment, a sophisticated attacker running a VM on the same physical host could potentially extract sensitive data — cryptographic keys, passwords, application secrets — from your workload's memory.
- **You cannot choose your neighbors:** In a public cloud, you generally have no control over which other tenants share your physical hardware.
- **Performance degradation from mitigations:** Security patches for side-channel attacks introduce CPU overhead. You pay for the same instance but get fewer useful cycles.
- **The vulnerability class continues to evolve:** Since 2018, researchers have discovered numerous additional speculative execution vulnerabilities — Foreshadow (L1 Terminal Fault), MDS (Microarchitectural Data Sampling), Zombieload, RIDL, Fallout, and others.
- **Mitigation strategies for your workloads:**
  - **Dedicated/Isolated Hosts:** AWS Dedicated Hosts, GCP Sole-Tenant Nodes, and Azure Dedicated Hosts guarantee that no other customer's workloads run on your physical hardware. Higher cost, but eliminates cross-tenant risk.
  - **Confidential Computing:** AMD SEV, Intel SGX, and Intel TDX encrypt VM memory so that even the hypervisor and host OS cannot read it. Available as Confidential VMs on all three providers.
  - **Keep guest OS and firmware updated:** Cloud providers patch the hypervisor, but customers are responsible for patching their guest operating systems and applications.
  - **Minimize sensitive data in memory:** Use encryption at rest and in transit, and consider using hardware security modules (HSMs) for cryptographic key management.
  - **Monitor for new disclosures:** Subscribe to cloud provider security bulletins.

Summary

Wrap-Up and Key Takeaways

- Virtualization is not cloud computing, but it is the technology that makes cloud computing possible. Without it, there is no multi-tenancy, no elasticity, and no pay-per-use.
- All three major cloud providers have converged on a similar architecture: minimal KVM-based hypervisors with custom hardware offloading networking, storage, and security to dedicated silicon.
- Choosing the right instance type, pricing model, and tenancy option is a core cloud engineering skill — it directly impacts cost, performance, and security.
- Machine images (AMIs) enable repeatability and automation — they are the bridge between infrastructure as code and running compute workloads.
- Hardware-level vulnerabilities like Meltdown and Spectre represent a fundamentally different class of security threat — they challenge the core trust model of shared infrastructure.
- As a cloud practitioner, understanding the shared responsibility model means knowing that the provider patches the hypervisor, but you patch your guests, your applications, and your architecture.

Additional Resources

See course lecture page on Canvas for supplementary materials and documentation links.

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